Training-Free Generation of Protein Sequences from Small Family Alignments via Stochastic Attention

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Generating novel protein sequences that respect a family's statistical constraints typically requires training deep generative models on thousands to millions of examples. Yet most protein families are small: the median Pfam seed alignment contains only 22 sequences, a regime where learned models overfit or collapse. We propose *stochastic attention* (SA), a training-free sampler that treats the modern Hopfield energy over stored sequences as a Boltzmann distribution and draws samples via Langevin dynamics. The score function is the residual of a single softmax attention operation, eliminating the need for a trained score network, pretraining data, or GPUs. Across eight Pfam families spanning 37 to 420 sequences and 23 to 262 residues, SA generates sequences with low composition divergence, substantial novelty, and structural plausibility supported by ESMFold and AlphaFold2. Compared with profile HMMs, EvoDiff, and the MSA Transformer, SA was the only tested method to simultaneously achieve low composition divergence, genuine novelty, and sequence identity within each family's nearest-neighbor identity range; the others drifted outside this range or produced near-copies. The critical inverse temperature is predicted from principal component analysis (PCA) dimensionality alone, enabling fully automatic operation from a seed alignment. In two domains with deep mutational scanning data, SA-generated substitutions are enriched for experimentally tolerated mutations beyond a position-matched null, and an independent language model (ESM2-650M) scores them within the natural range. Stochastic attention thus opens training-free sequence generation to the long tail of protein families too small for deep learning.

protein sequence generation | stochastic attention | Hopfield networks | Langevin dynamics | training-free generative models

With single-chain protein structure prediction transformed by AlphaFold2 (1) and RoseTTAFold (2), the frontier has shifted toward *designing* novel protein sequences with desired properties (3, 4). Generative approaches to address this problem now span a broad landscape, from classical profile hidden Markov models (pHMMs) (5, 6) and Potts models (7–9) to deep generative models including variational autoencoders (10, 11), autoregressive language models (12, 13), alignment-processing transformers (14), MSA-conditioned diffusion (15), and structure-conditioned methods such as ProteinMPNN (16) and RFdiffusion (17). Yet all sequence-level approaches share a common requirement: large training corpora. Language models are pretrained on millions of UniRef50 sequences; even MSA-conditioned methods require either extensive pretraining or large input alignments. This creates a gap for small families: the median Pfam seed alignment, the curated set of representative sequences that alignment-based methods take as input, contains only 22 sequences, and nearly half of all families have fewer than 20 sequences (SI Appendix, Section S23), well below the minimum viable training set for any deep generative architecture. In this scarce-data regime, deep models overfit and collapse in diversity, while pHMMs can emit sequences with low identity to the source family. What is needed is an approach that can extract the statistical structure of a protein family from whatever data is available, even a few dozen sequences, without learning parameters that could lead to overfitting.

The connection between attention and associative memory supplies exactly this. Ramsauer et al. (18) showed that a single softmax attention operation is equivalent to one step of the retrieval dynamics on the modern Hopfield energy, building on the dense associative memory framework of Krotov and Hopfield (19). This connection endows any set of stored examples with a well-defined energy landscape: the examples become memory patterns, and the Boltzmann distribution over this landscape assigns high probability to states that resemble the stored set. The energy itself has a transparent structure: a quadratic term pulls a candidate state toward the origin, while a competing log-sum-exp term over its similarities to the stored patterns pulls it toward whichever memories it most resembles, with an inverse temperature controlling how sharply the nearest pattern dominates. The gradient of the log-sum-exp is precisely the softmax-weighted combination of stored patterns, i.e., a single attention operation. The score function of the Boltzmann distribution, the gradient that guides sampling toward higher-probability states, is therefore the *residual* of attention: the difference between the attention output and the current state. Diffusion-based generative models such as RFdiffusion (17) obtain such a gradient by training a neural network on data; here it is instead exact and available in closed form, so Langevin sampling can proceed without learning any parameters or approximating gradients. We exploit this insight to propose *stochastic attention* (SA), a training-free protein sequence generator that combines the exact Hopfield score function with Langevin dynamics. Our prior work (20) introduced SA and validated it on synthetic data, MNIST, and a single protein family. The present study advances substantially beyond that proof of concept, moving from a single illustrative family to a systematic evaluation across diverse families, with the scaling analysis, baseline benchmarking, and biological validation that the earlier work did not attempt.

In this study, we explored whether stochastic attention (SA) can serve as a practical generative model for small protein families. We systematically evaluated SA across eight diverse Pfam families spanning a ten-fold range of family sizes and a seven-fold range of sequence lengths, assessing compositional fidelity, sequence novelty, and structural plausibility using both ESMFold and AlphaFold2. We characterized how generation quality scaled with family size, uncovering an empirical law that predicts the critical temperature from alignment statistics alone and eliminates the need for a temperature sweep. We tested robustness in a scarce-data regime down to $K=20$ sequences, compared SA head-to-head against four established baselines (profile HMMs, EvoDiff, MSA Transformer, and a Potts model), and examined whether the generated sequences preserved the pairwise residue covariation that reflects structural contacts and functional couplings. Finally, we sought independent validation of biological plausibility through ESM2-650M pseudo-perplexity scoring and cross-referencing against published deep mutational scanning datasets. Across all eight families, SA generated novel sequences that preserved family-level amino acid statistics and were predicted by both ESMFold and AlphaFold2 to adopt the target-family fold, a combination not matched by the learned baselines tested here. Together these results indicate that the statistical structure of a protein family can itself support sequence generation, without iterative parameter optimization, backpropagation, or external pretraining, when the sampler respects the geometry of the family's sequence space.

Results

We ran SA on all eight Pfam families (Table 1) and observed that the method consistently generated novel protein sequences while preserving family-level amino acid statistics, across a ten-fold range of family sizes and a seven-fold range of sequence lengths (Fig. 1; SI Appendix, Table S1). We first assessed sampler accuracy by comparing the outputs of the unadjusted Langevin algorithm (ULA) and its Metropolis-adjusted

Significance Statement

Most protein families are small, with a median of only 22 sequences in their Pfam seed alignments, too few to train a deep generative model. We show that a single attention operation over a family alignment defines an energy landscape whose Boltzmann distribution can be sampled by Langevin dynamics without model training, GPUs, or external data. The resulting method, stochastic attention, generates novel protein sequences that preserve family-level amino acid statistics and are predicted by independent structure predictors to adopt the target-family fold across eight families. A linear relationship between representation dimensionality and critical temperature enables fully automatic operation from a seed alignment, opening training-free generation to the many protein families too small for deep learning.

J.D.V. designed research, performed research, analyzed data, and wrote the paper.

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variant (MALA): acceptance rates exceeded 99.6% in every family, so the Metropolis correction rejected fewer than 0.4% of proposed moves at this step size (SI Appendix, Table S11); we treat this as evidence that the unadjusted and adjusted updates rarely differ in this regime, and adopt the computationally cheaper ULA in all subsequent experiments. Generated sequences were not near-duplicates of stored patterns: zero sequences across all eight families exceeded 90% identity to any stored sequence (SI Appendix, Table S13), and the generated ensemble was insensitive to initialization strategy (SI Appendix, Table S12). In the generation regime, the sampler operated at inverse temperature $\beta \approx 2\beta^*$, where the critical inverse temperature β^* is the value at which the energy landscape changes character: below β^* the stored patterns blur into a single broad well, while above it the landscape resolves into distinct basins around individual patterns (SI Appendix, Section S21 gives the criterion that locates it). At this setting, the generated amino acid composition closely matched each natural family's: the Kullback–Leibler (KL) divergence between them, which falls to zero only when two distributions are identical, stayed below 0.06 in every family and below 0.02 in several. Novelty ranged from 0.40 (Defensin_beta) to 0.65 (WW), indicating substantial departure from any single stored memory, and all 150 generated sequences were distinct in every family, with mean pairwise sequence identity of 35–65% (equivalently, mean pairwise Hamming distance 0.36–0.65; SI Appendix, Table S14), indicating that the sampler produced a diverse ensemble rather than collapsing to a single mode. Nearest sequence identity to a stored pattern was 51–66%, within each family's empirical within-family nearest-neighbor identity envelope (each natural sequence's identity to its closest neighbor in the same family; SI Appendix, Table S14) in seven of eight families; in Pkinase, generated identity (0.515) slightly exceeded the natural maximum (0.454), consistent with that family's strongly consensus-proximal generation. Generated sequences thus remained within the observed sequence-similarity envelope while departing meaningfully from stored patterns. By contrast, bootstrap samples had zero novelty by construction and Gaussian perturbation produced negligible novelty (< 0.02), indicating that small additive noise is insufficient to escape the basin of the nearest stored pattern.

Position-specific analysis of the Kunitz domain revealed why SA preserves family statistics (SI Appendix, Fig. S5). Sequences generated by stochastic attention preserved all 11 positions with greater than 90% conservation in the seed alignment, including all six cysteines forming the three disulfide bonds essential to the Kunitz fold, while introducing 17–25 substitutions concentrated at variable positions. Per-position Shannon entropy correlated strongly between the SA-generated ensemble and the stored alignment (Pearson r=0.968), indicating that the method identified which positions tolerate variation and which do not. This behavior followed directly from the Boltzmann sampling mechanism: at strongly conserved positions, all stored patterns agreed and the softmax weights concentrated probability on the consensus residue; at variable positions, stored patterns diverged and the sampler drew from the local distribution of tolerated amino acids.

Subsampling the WW domain at $K \in \{20, 50, 100, 200, 400\}$ with five independent replicates per condition characterized how generation quality changed with decreasing family size (SI Appendix, Fig. S2). Stochastic attention maintained low KL divergence across the entire range (0.019 ± 0.008 at K=20, improving to 0.010 ± 0.002 at K=400), while the PCA dimensionality grew from d=17–18 components at K=20 to d=182–183 at K=400 and the critical temperature tracked this growth (β^*=2.7 to 5.5). Despite the roughly ten-fold increase in effective dimensionality, SA generation quality remained stable, suggesting that the entropy inflection criterion successfully adapted the operating temperature to the geometry of the memory matrix at each scale. Novelty increased with K (from 0.47 to 0.74), reflecting the richer landscape of a larger memory set, while nearest sequence identity decreased correspondingly (from 0.71 to 0.56), indicating that the sampler explored further from stored patterns when more patterns were available to collectively shape the energy landscape. Replication across three additional families at K=20 (SH3, Kunitz, zf-C2H2) confirmed that this robustness is not family-specific (SI Appendix, Table S3).

The scaling study revealed a consistent pattern in how the sampler self-tunes: the more independent variation a family contained, the higher the inverse temperature the sampler selected for it. Concretely, the critical inverse temperature β^* tracked the family's PCA dimensionality d, the number of principal components needed to capture its sequence variation, which prompted us to ask whether β^* could be predicted from alignment statistics alone, without a temperature sweep (Fig. 2). Fitting the eight independent Pfam families alone yielded a simple two-parameter model that explained 95% of the variance (R^2=0.95), and leave-one-family-out cross-validation confirmed out-of-sample generalization (SI Appendix, Section S21); because this fit rests only on independent families, we treat it as the primary evidence. Pooling all 33 data points (the 8 families and 25 WW scaling replicates) gave a nearly identical relationship, $\beta^* \approx 1.52 + 0.28\sqrt{d}$ (bootstrap standard error [SE]: intercept ±0.07, slope ±0.007; R^2=0.97), which we report as supporting evidence because 25 of the 33 points are nested WW subsamples and therefore not independent. The fitted slope of 0.28 falls well below the $\sqrt{d}$ coefficient of ≈1.6 predicted by a Gaussian random-score model. This reduction arises from two effects: concentration of measure on the unit sphere $\mathbb{S}^{d-1}$ sets the random-probe similarity scale at $\mathcal{O}(1/\sqrt{d})$, while self-similarity anchoring in the stored-pattern probes (each stored pattern has unit inner product with itself, creating a gap of 0.55–0.82 over the next-nearest pattern) shifts the inflection to lower β than a Gaussian mean-field model predicts; together these effects reflect the structured similarity spectrum of real protein sequences arising from conserved positions, correlated substitutions, and phylogenetic signal (SI Appendix, Section S21). Adding further predictors (mean column entropy, effective number of sequences, spectral concentration) produced negligible improvement ($\Delta R^2 < 0.001$), indicating that PCA dimensionality captures most of the relevant variation. The empirical β^* is identified on a fixed 50-point logarithmic grid over $[0.1, 500]$ (multiplicative spacing ≈1.19), so it is resolved to within ≈19% and the reported R^2 is computed against these grid-resolved targets. This regression provides a practical shortcut: given a new family, one can estimate β^* from PCA alone and set the generation temperature without a sweep.

We compared SA against three established learned baselines, profile HMM emit, EvoDiff (MSA-conditioned diffusion), and the MSA Transformer (Gibbs masked language model sampling), across all eight families (Fig. 3). Reading the three panels jointly is essential: no single metric tells the full story, and high novelty is only useful when accompanied by compositional fidelity and sequence identity consistent with the target family. Profile HMM emit and the MSA Transformer achieved high novelty scores, but their nearest sequence identity to stored patterns fell to 9–59% in several families. The relevant natural reference is the within-family nearest-neighbor identity, that is, each natural sequence's identity to its closest neighbor in the same family, which is the same quantity reported for the generated sequences (and is necessarily higher than the mean pairwise identity of 22–37%, which averages over all distant pairs). In the families where the baselines drifted, their nearest sequence identity fell below the minimum nearest-neighbor identity of any natural sequence: profile HMM emit reached 11–21% in Pkinase, PDZ, and RRM, whereas the lowest nearest-neighbor identity among natural sequences in those families is 22–28% (SI Appendix, Table S14). The green boxes in Fig. 3C mark each family's natural within-family nearest-neighbor identity envelope; a sequence falling below its family's box may have moved outside the family, whereas one rising above it is a near-copy of stored patterns with insufficient novelty. Thus, the high novelty of HMM emit and the MSA Transformer was accompanied by sequence-identity drift rather than clearly family-consistent exploration. EvoDiff balanced the metrics more effectively in shorter families but required pretraining on millions of UniRef50 MSAs, scaled poorly with sequence length (∼14 hours for 150 Pkinase sequences on CPU, though GPU inference would reduce this substantially, versus seconds for SA on the same hardware; SI Appendix, Table S15), and on the longest family generated sequences well short of the alignment length, which after alanine padding registered as elevated composition divergence (Pkinase KL=0.17, more than an order of magnitude above its values in the other seven families).

Among the tested methods, stochastic attention was the only one that simultaneously achieved all three criteria (Fig. 3). It held KL divergence below 0.05 in seven of eight families, produced novelty of 0.40–0.65 indicating departure from stored patterns (zero generated sequences exceeded 90% identity to any stored pattern, and zero were within two substitutions of a stored sequence across all eight families), and kept sequence identity at 51–66%, within each family's empirical within-family nearest-neighbor identity envelope for seven of eight families, with Pkinase marginally above (SI Appendix, Table S14). This combination (compositionally faithful, novel, and within the family's sequence-similarity envelope) distinguished SA from the baselines tested, using only the seed alignment and no pretraining, backpropagation, or GPU resources. A systematic comparison with a Potts model fitted via pseudo-likelihood maximization (plmDCA) across all eight families (SI Appendix, Table S10) provided a more informative contrast, because Potts models are a natural physics-based competitor: they explicitly fit pairwise coupling parameters $J_{ij}(a, b)$ between all position pairs, giving them direct access to coevolutionary signal. The comparison revealed comparable compositional fidelity and broadly similar output quality in six of eight families, with SA achieving higher novelty in five of eight. However, the Potts model showed limitations in the most data-starved families (Pkinase: novelty of 0.22 vs. 0.48 for SA, indicating the Gibbs sampler remained near stored patterns; Defensin_beta: KL=0.112, the highest of any method), consistent with its high parameter count: the number of coupling parameters scales as $L(L-1)q^2/2$ for alignment length L and alphabet size q=20, yielding data-to-parameter ratios as low as 2.5×10^{-6}. That SA achieves comparable or better output without fitting any parameters suggests that the Hopfield energy implicitly captures some of the correlational structure that pairwise coevolutionary models fit explicitly, but through the softmax attention operation rather than through parameter estimation. Whereas the Potts model encodes pairwise couplings, the softmax nonlinearity in the Hopfield energy couples all positions simultaneously through the attention weights, giving it the capacity to encode higher-order correlations without the combinatorial growth of explicit higher-order terms.

Two control experiments indicated that SA captured more than marginal per-position amino acid frequencies (SI Appendix, Tables S16 and S17). First, a consensus-with-noise baseline that added calibrated isotropic Gaussian perturbations in PCA space produced a qualitatively different output profile: systematically higher novelty (0.63–0.79 vs. 0.40–0.65 for SA) paired with lower sequence identity to stored patterns (0.44–0.62 vs. 0.52–0.66), reflecting the fact that isotropic noise scattered sequences away from all stored patterns rather than interpolating between them in a structured way. Second, SA run on column-permuted alignments, which preserved the exact per-position amino acid frequencies but destroyed all inter-position correlations, yielded consistently different output: higher novelty and lower sequence identity than SA on the real alignment in all eight families, with MI correlation differences favoring the real alignment in seven of eight families. The larger differences in novelty and sequence identity indicate that inter-position correlations present in real alignments constrain the energy landscape in ways that marginal frequencies alone cannot reproduce, and that SA exploits this correlational structure during generation in most families.

The computational cost of SA differed from learned baselines by orders of magnitude (SI Appendix, Table S15). Stochastic attention generated 150 sequences in 0.2–4.6 seconds across all eight families on a single CPU core, with the variation driven primarily by the number of stored patterns K (which determines the cost of the softmax attention at each Langevin step). Profile HMMs were consistently the fastest method (0.02–0.09 seconds), as expected for sampling from a precomputed position-specific model. EvoDiff required 2–14 hours on CPU depending on sequence length, and the MSA Transformer required 1–8 hours for iterative masked language model sampling. Potts model fitting and Gibbs sampling required 1–52 minutes, with the Pkinase family (L=262) dominating due to the $\mathcal{O}(L^2q^2)$ parameter count. These comparisons place SA in a different computational regime from the learned baselines tested here: it is 3–5 orders of magnitude faster than EvoDiff and the MSA Transformer in these CPU-only runs, requires no GPU, no pretraining data, and no parameter optimization. These figures reflect the CPU-only, fixed-length evaluation protocol used here rather than a general method-level speed ranking: GPU inference would substantially reduce the EvoDiff and MSA Transformer times, and the alanine padding and truncation used to conform variable-length outputs to the alignment length penalize those generators on the longest families. This accessibility follows from the closed-form score function, which eliminates the need for backpropagation, gradient accumulation, or iterative parameter updates.

We assessed structural plausibility by predicting folded structures for 50 sequences per method per family using ESMFold (Fig. 4), comparing SA generation to stored natural sequences via Wilcoxon rank-sum tests with Cohen's d effect sizes. Sequences generated by stochastic attention achieved mean predicted local distance difference test (pLDDT) scores, the predictor's own per-residue confidence that each position is placed correctly (reported from 0 to 100), statistically indistinguishable from stored sequences in six of eight families ($p > 0.05$), with significantly higher pLDDT in Kunitz (90.8 vs. 89.3, p=0.001, d=0.67), and at least 86% of SA-generated sequences exceeded the pLDDT=70 confidence threshold in seven of eight families, with SH3 the lone exception (54%). The template modeling (TM) score, which runs from 0 to 1 and rises as a predicted structure superimposes more closely on the reference, showed a consistent pattern: in six of eight families, the predicted structures of SA-generated sequences matched the reference significantly better than those of stored sequences did ($p < 0.05$; $d > 1.0$ in RRM, WW, PDZ, and Pkinase). Two explanations merit consideration (predictor preference for consensus-like sequences, or genuine structural convergence of the Boltzmann ensemble toward the shared fold), which we weigh in the Discussion. As a direct test of the first, we asked whether predicted structural quality tracks consensus proximity within each family: it does so only weakly and inconsistently (per-family Spearman ρ between TM-score and identity to the consensus ranged from -0.13 to 0.53, pooled ρ=0.17; pLDDT pooled ρ=0.23; SI Appendix, Section S6), indicating that the elevated TM-scores are not primarily an artifact of consensus proximity. As an independent cross-validation, we repeated the structure prediction for all eight families using AlphaFold2 (SI Appendix, Figs. S3–S4). The AlphaFold2 results corroborated the ESMFold findings while resolving the single negative result: for SH3, the only family where ESMFold had indicated a pLDDT deficit, AlphaFold2 showed no pLDDT difference (p=0.99) and significantly higher TM-scores for SA generation (0.736 vs. 0.721, $p < 0.001$). The concordance between two independent structure predictors across all eight families strengthens the computational evidence that SA-generated sequences are predicted to adopt the target-family fold (Fig. 5).

Pairwise mutual information (MI) analysis tested whether SA preserved residue covariation, which reflects structural contacts and functional couplings that single-site statistics cannot capture (SI Appendix, Table S6; Figs. S6–S7). Pearson correlations between stored and SA-generated MI vectors ranged from r=0.53 to 0.92 across seven of eight families, with Pkinase (r=0.05) the lone exception. This exception has a specific cause. The Pkinase alignment is shallow, only K=37 sequences over L=262 positions, which could in principle make its covariation hard to estimate; but that is not the explanation, because the Potts model recovers the covariation on the identical alignment (r=0.95). The failure lies instead in SA generation, which in Pkinase stays closer to the family consensus than in any other family (mean identity to the consensus 0.73, the highest observed; SI Appendix, Section S6); because near-consensus sequences vary little from one another, they carry almost none of the position-to-position covariation that MI measures. Stochastic attention thus preserved covariation in the compact families but not in the longest, most data-starved one. Stochastic attention exceeded profile HMMs (whose MI correlations spanned r=0.07–0.83) in seven of eight families, and the general ordering HMM $<$ SA $<$ Potts $\leq$ EvoDiff held in three of eight families, consistent with the information hierarchy of each approach: position-independent frequencies, implicit all-order Hopfield couplings, explicitly fitted pairwise couplings, and deep coevolutionary priors from large-scale pretraining, respectively. This ordering supports the view that the Hopfield energy captured correlational structure beyond what position-independent models encode, without requiring explicit pairwise parameter estimation.

As a complementary, sequence-level assessment of biological plausibility, we scored all SA-generated and stored sequences using ESM2-650M (21), a protein language model pretrained on millions of UniRef50 sequences that is entirely independent of SA (SI Appendix, Table S7). Sequences generated by stochastic attention were statistically indistinguishable from natural family members in four of eight families (RRM, zf-C2H2, Pkinase, Defensin_beta; Wilcoxon $p > 0.29$), with SA achieving significantly *better* pseudo-perplexity in the Kunitz domain (4.47 vs. 4.98, $p < 0.001$, Cohen's d=-0.74). Three families showed modestly higher pseudo-perplexity for SA generation (SH3, WW, PDZ; d=0.60–1.03), though all SA-generated sequences remained well within the range of natural proteins. Across all eight families, the mean pseudo-perplexity was 5.71 for SA-generated sequences versus 5.58 for stored sequences, indicating that an independent model trained on billions of natural sequences scores SA-generated sequences comparably to natural family members. Beyond this language-model scoring, we cross-referenced SA-generated substitutions against published deep mutational scanning (DMS) datasets to ask whether the Boltzmann sampling procedure preferentially generates mutations that are individually tolerated in experimental assays (SI Appendix, Tables S7 and S8). For the PDZ domain (DLG4/PSD-95 PDZ3 (22)), 91.7% of 4,313 matched substitutions were classified as experimentally tolerated, and for the SH3 domain (GRB2 SH3 (23)), 90.4% of 2,396 matched substitutions were tolerated. Because SA preferentially substitutes at variable, more tolerant positions, comparing these rates against the overall dataset tolerance rate (77.5% and 67.5%) conflates position selection with residue choice. We therefore used a position-matched null that asks, for each position SA substituted, how often a randomly chosen tested substitution at that same position is tolerated. Against this stricter null, SA-generated substitutions remained significantly enriched in both families (PDZ: 91.7% vs. 84.3%, 1.09×, $p < 10^{-77}$; SH3: 90.4% vs. 74.9%, 1.21×, $p < 10^{-121}$; a conservation-matched null gives the same result; SI Appendix, Table S9), indicating that SA selects function-compatible residues beyond simply targeting tolerant positions. The continuous DMS fitness scores of SA-generated substitutions were likewise significantly higher than those of random single mutations in both families (Wilcoxon $p < 10^{-20}$; SI Appendix, Tables S7 and S8). These results indicate that the Boltzmann sampling procedure preferentially generates substitutions at positions and to residues that are individually compatible with function in DMS assays, without any explicit fitness objective; they do not by themselves establish that every full generated sequence is functional, and the DMS evidence is limited to these two domains.

Finally, the beta defensin family (PF00711) provided an opportunity to test whether SA generation preserves not only sequence-level statistics but also the biophysical properties that underpin antimicrobial function (SI Appendix, Fig. S8). Sequences generated by stochastic attention preserved the six-cysteine disulfide scaffold (6.01 ± 0.08 mean cysteines, compared to 6.02 ± 0.15 for natural sequences), maintained a cationic charge profile ($+6.1 \pm 1.6$) with the narrowest charge distribution of any method, and occupied the same region of charge-hydrophobicity space as natural defensins. By a minimal antimicrobial plausibility filter (net charge $\geq +2$, hydrophobic fraction in $[0.3, 0.7]$, ≥ 4 cysteines), 51% of SA-generated sequences passed, compared to 42% of stored sequences, 25% of HMM-emitted sequences, and 16% of MSA Transformer sequences. This implicit enforcement of biophysical constraints arises because the energy landscape concentrates probability near the stored patterns, which already satisfy these constraints by virtue of being functional proteins; no explicit objective for charge, hydrophobicity, or disulfide topology is required.

Discussion

Across eight Pfam families spanning a ten-fold range of sizes (K=37–420) and a seven-fold range of sequence lengths (L=23–262), SA generated novel sequences that preserved family-level amino acid statistics and were predicted by independent structure predictors to adopt the target-family fold, using only the seed alignment and no training, external data, or GPU resources. None of the learned baselines tested here achieved this combination. Profile HMMs and the MSA Transformer generated sequences with nearest-neighbor identity below the empirical within-family nearest-neighbor envelope, in several families below the minimum observed among natural sequences (SI Appendix, Table S14), indicating that their high novelty was accompanied by sequence-identity drift rather than family-consistent exploration. EvoDiff required millions of pretraining sequences and hours of compute, and degraded on the longest family. Among the tested baselines, no method simultaneously stayed within the family identity range, maintained low composition divergence, and produced genuine novelty. That a training-free method matched systems built on orders of magnitude more data indicated that the statistical structure of protein families can support generation without iterative parameter optimization, backpropagation, or external pretraining, provided the generative framework respects the geometry of the family's sequence space. Beyond matching their output, SA avoids the pretraining corpora, GPU resources, and large parameter counts that place deep generative methods out of reach for the small, single-laboratory families that make up most of Pfam, requiring only a seed alignment and a standard workstation. The consensus-with-noise and column-permuted alignment controls supported the conclusion that this was not only a consequence of marginal statistics: SA produced a different output profile from isotropic perturbation of the consensus, and exploited inter-position correlations that column-independent models could not capture.

Sequences generated by stochastic attention were predicted to adopt the canonical family fold, and in six of eight families their TM-scores to the reference structure modestly exceeded those of the stored natural sequences under ESMFold (AlphaFold2 reproduced the pattern across all eight families, the two predictors' TM-scores correlating at r=0.999; SI Appendix, Fig. S4). The pLDDT scores were indistinguishable from stored sequences in six of eight families, supporting the conclusion that the sampler did not sacrifice predicted folding confidence for diversity, and the single exception (SH3) was resolved by AlphaFold2 cross-validation. Two explanations for the elevated TM-scores merit consideration. First, predictor bias: structure predictors trained on evolutionary data may assign higher confidence to consensus-proximal sequences. Second, genuine structural convergence: the Boltzmann ensemble averages over lineage-specific deviations, potentially producing sequences that encode the shared fold more cleanly than individual natural sequences. A direct test of the first explanation is whether predicted structural quality tracks consensus proximity within each family; it does so only weakly and inconsistently (pooled Spearman ρ=0.17 for TM-score and 0.23 for pLDDT, each accounting for under 4% of the variance, and negative in three families; SI Appendix, Section S6), which argues against predictor bias as the sole driver, though it does not exclude an absolute preference for family-typical sequences shared by both the sampler and the predictor. We therefore interpret the elevated TM-scores cautiously and do not read them as evidence that SA-generated sequences fold better than natural ones: the two explanations are not mutually exclusive, and the decisive computational control, folding the consensus-with-noise ensemble through the same structure predictors, was not performed here and remains for future work, while definitive attribution requires experimental structure determination. The DMS validation provided partial evidence for the latter: if elevated TM-scores reflected only predictor bias, one would not necessarily expect SA substitutions to be enriched for experimentally tolerated mutations, yet the position-matched enrichment of 1.09–1.21× in two independent DMS datasets suggested that the generated substitutions are biased toward locally function-compatible changes. However, all computational validators used here (structure predictors, ESM2-650M, and DMS-derived fitness scores) were trained on evolutionary data, so high scores for family-like sequences are expected; experimental characterization would provide definitive confirmation.

The WW domain scaling study showed that generation quality was stable down to K=20 sequences, with KL divergence below 0.02 and novelty above 0.47, and replication across three additional families supported the same trend. This robustness arose because the energy landscape is defined entirely by the stored patterns, with no parameters that could overfit to a small training set. The entropy inflection criterion adapted the operating temperature to the memory matrix at each scale, and β^* could be predicted from PCA dimensionality alone (R^2=0.97), enabling fully automatic operation from a seed alignment. The $\sqrt{d}$ scaling of β^* is consistent with concentration of measure on the unit sphere: random-probe similarities have variance $1/d$, so the inverse temperature required to resolve $\mathcal{O}(1/\sqrt{d})$ similarity differences grows as $\sqrt{d}$. The reduced coefficient (0.28 vs. $\sim$1.6 for random patterns) reflected self-similarity anchoring in the stored-pattern probes: each stored pattern has unit inner product with itself but substantially lower similarity to other patterns, so the softmax concentrated on the self-pattern at lower β than a mean-field model predicts (SI Appendix, Section S21). By contrast, learned generative models face a parameter-to-data ratio problem when $K < 100$: VAEs and diffusion models have thousands to millions of parameters that cannot be constrained by a few dozen sequences. This regime encompasses most Pfam families.

The connection between attention and Boltzmann sampling extends beyond proteins: any set of examples representable as vectors defines an energy landscape via the modern Hopfield energy, and Langevin dynamics on this landscape provides a generative model with no training. Two conditions determine whether the approach produces useful output. First, the stored examples must be structured enough that the energy landscape has well-defined basins rather than a featureless plateau; protein families satisfy this because evolutionary constraint creates correlated patterns of conservation and variation. Second, the examples must be diverse enough that the sampler does not trivially retrieve: if all stored patterns are nearly identical, the distribution collapses and generation reduces to consensus with noise. The eight families spanned 22–37% mean pairwise identity, diverse enough to avoid consensus collapse yet, through their correlated conservation and variation, structured enough to form well-defined basins. Classical Hopfield networks were limited to storing $\mathcal{O}(d)$ patterns before catastrophic interference (24, 25). The modern continuous Hopfield energy lifts this limit, storing a number of well-separated patterns that grows exponentially with d (18). This is an advantage for scarce-data generation: with $K \ll d$ stored patterns the family lies far below capacity, so the energy landscape is well-separated and the sampler can explore between memories without interference. The same approach may apply to other biological sequence families where data is scarce but examples are structured, such as antibody repertoires where clonotypes have only a handful of somatic variants, RNA families with few seed sequences, or small-molecule libraries with tens of analogs per series. The Hopfield energy also admits a conditioning mechanism: adjusting pattern multiplicities via a scalar bias on the attention logits shifts generation toward a user-specified functional subset, enabling property-directed generation (e.g., toward binding activity) without retraining (26). The analytic score function makes the sampler amenable to the convergence theory of Langevin dynamics (27, 28): in the strongly regularized, log-concave regime ($\beta\sigma^2_{\max} < 2$, with $\sigma_{\max}$ the largest singular value of the memory matrix) these results yield formal guarantees on sample quality that are absent from most neural generative models. The generation temperatures used here lie outside that regime ($\beta\sigma^2_{\max} \gg 2$; SI Appendix, Section S22), where the energy is non-convex and convergence is supported by the empirical mixing diagnostics and initialization-insensitivity analysis rather than by these bounds.

Several limitations of the SA approach should be acknowledged. The most significant is that SA operates on a fixed alignment representation: it requires a pre-computed MSA and a fixed alignment length L, and cannot generate insertions, deletions, or variable-length sequences. This constraint is inherited from the one-hot-plus-PCA encoding, which maps each sequence to a vector of fixed dimension $20L$; any change in L would alter the representation space itself. In practice, this means the method is best suited to compact, well-aligned domains rather than multi-domain proteins or families with frequent large insertions. Extending stochastic attention to variable-length generation would likely require replacing PCA with a length-agnostic embedding while preserving the closed-form score function, an open direction for future work. The PCA projection itself discards information that may be relevant for capturing higher-order correlations, though a sensitivity analysis confirmed robustness across the 75%–99% retained variance range (SI Appendix, Table S18). Relatedly, SA preserved pairwise residue covariation in seven of eight families but largely lost it in Pkinase (r=0.05), the longest and most data-starved family, where generation was strongly consensus-proximal (SI Appendix, Table S6); covariation fidelity therefore degrades when the data-to-length ratio is very low, and our covariation claims are restricted to the compact families. Further, the eight families studied here are all well-structured, globular domains with abundant structural and functional annotation; this choice was necessary for validation but introduced a selection bias toward precisely the class of proteins that structure predictors handle best. Performance on intrinsically disordered regions, transmembrane domains, multi-domain proteins, or families with very low sequence identity remains untested, and the method's reliance on PCA may be particularly limiting for disordered families where sequence variation is less structured. Finally, our multi-chain protocol does not guarantee coverage of all modes in the Boltzmann distribution, particularly for very large or highly diverse families, and wet-lab characterization of selected generated sequences would be needed to establish the method's utility for protein engineering applications.

Materials and Methods

Stochastic Attention. Let $\mathbf{X} = [\mathbf{m}_1, \ldots, \mathbf{m}_K] \in \mathbb{R}^{d\times K}$ be the memory matrix whose columns are vectorized protein sequences from a family alignment of K members, each represented in d dimensions after one-hot encoding and PCA projection. Each sequence of length L is first one-hot encoded to a binary vector in $\{0,1\}^{20L}$, then projected by PCA (retaining 95% of variance, yielding $d \ll 20L$) and unit-normalized to form the corresponding column $\mathbf{m}_k$; full encoding, projection, and decoding details are given in SI Appendix, Section S20. The modern Hopfield energy (18) of a state $\boldsymbol{\xi} \in \mathbb{R}^d$ is:

$$E(\boldsymbol{\xi}) = \tfrac{1}{2}\|\boldsymbol{\xi}\|_2^2 - \tfrac{1}{\beta}\log\left(\sum_{k=1}^{K}\exp\!\big(\beta\,\mathbf{m}_k^\top\boldsymbol{\xi}\big)\right), \qquad [1]$$

where $\beta > 0$ is an inverse temperature. The Boltzmann distribution $p_\beta(\boldsymbol{\xi}) \propto \exp(-\beta\,E(\boldsymbol{\xi}))$ has score function $\nabla \log p_\beta(\boldsymbol{\xi}) = \beta(\mathbf{T}(\boldsymbol{\xi}) - \boldsymbol{\xi})$, where $\mathbf{T}(\boldsymbol{\xi}) := \mathbf{X}\,\mathrm{softmax}(\beta\,\mathbf{X}^\top\boldsymbol{\xi})$ is the softmax attention retrieval map. This score is *exact* and computed by a single attention operation; no score network or training is required. Applying the unadjusted Langevin algorithm (ULA) (29) to the potential $U = \beta E$ with step size α/β, where $\alpha \in (0, 1)$ is the mixing coefficient, yields the SA update:

$$\boldsymbol{\xi}_{t+1} = (1-\alpha)\,\boldsymbol{\xi}_t + \alpha\,\mathbf{X}\,\mathrm{softmax}\!\big(\beta\,\mathbf{X}^\top\boldsymbol{\xi}_t\big) + \sqrt{\tfrac{2\alpha}{\beta}}\,\boldsymbol{\epsilon}_t, \quad \boldsymbol{\epsilon}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \qquad [2]$$

Each step performs contraction toward the origin, a softmax-weighted pull toward stored memories, and isotropic Gaussian noise, at a per-step cost of $\mathcal{O}(dK)$. A complete derivation of this update, from the Gibbs form of the target density through the closed-form Hopfield score to the Langevin step, is given in SI Appendix, Section S1. The full generation procedure, from one-hot encoding and PCA projection through entropy-based temperature selection to multi-chain Langevin sampling and decoding, is given in Algorithm 1.

Algorithm 1 Stochastic Attention Protein Sequence Generation

Require: Seed alignment $\mathbf{S} \in \mathcal{A}^{K\times L}$, variance threshold $\rho_{\min} = 0.95$, mixing coefficient $\alpha = 0.01$, chains $N_c = 30$, iterations $T = 5000$, burn-in $T_b = 2000$, thinning $\Delta t = 100$, samples/chain $s = 5$

Ensure: Generated sequences $\{\mathbf{g}_1, \ldots, \mathbf{g}_S\} \subset \mathcal{A}^L$, $\quad S = N_c \cdot s$

Encoding:
One-hot encode: $\mathbf{B} \leftarrow [\mathrm{onehot}(\mathbf{S}_{1,:}), \ldots, \mathrm{onehot}(\mathbf{S}_{K,:})] \in \{0,1\}^{20L\times K}$
Center: $\bar{\mathbf{x}} \leftarrow \frac{1}{K}\sum_k \mathbf{B}_{:,k}$; $\quad \tilde{\mathbf{B}} \leftarrow \mathbf{B} - \bar{\mathbf{x}}\mathbf{1}_K^\top$
SVD: $\tilde{\mathbf{B}} = \mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^\top$
Select d: smallest p such that $\sum_{j=1}^{p}\sigma_j^2 / \sum_j \sigma_j^2 \geq \rho_{\min}$; set $\mathbf{W}_d \leftarrow \mathbf{U}_{:,1:d}$
Project and normalize: $\mathbf{m}_k \leftarrow \mathbf{W}_d^\top(\mathbf{B}_{:,k} - \bar{\mathbf{x}}) / \|\mathbf{W}_d^\top(\mathbf{B}_{:,k} - \bar{\mathbf{x}})\|_2$ for $k = 1, \ldots, K$
Assemble memory: $\mathbf{X} \leftarrow [\mathbf{m}_1, \ldots, \mathbf{m}_K] \in \mathbb{R}^{d\times K}$
Temperature selection:
Set $K_p \leftarrow \min(K, 20)$ and evaluate 50 log-spaced β values in $[0.1, 500]$
For $i = 1, \ldots, K_p$, compute $\mathbf{p}_i(\beta) = \mathrm{softmax}(\beta\,\mathbf{X}^\top\mathbf{m}_i)$ and $H_i(\beta) = -\sum_{k=1}^{K} p_{ik}(\beta)\log p_{ik}(\beta)$
Set β^* to the entropy inflection point of $\bar{H}(\beta) = K_p^{-1}\sum_i H_i(\beta)$ on this grid
Set $\beta_{\text{gen}} \leftarrow \max\!\big(\lceil 2\beta^* \rceil,\ 5\big)$
Multi-chain Langevin sampling:
$\mathcal{G} \leftarrow \emptyset$
for $c = 1, \ldots, N_c$ **do**
    Draw $k_c \sim \mathrm{Uniform}\{1, \ldots, K\}$
    $\boldsymbol{\xi}_0 \leftarrow \mathbf{m}_{k_c} + 0.01\,\boldsymbol{\eta}$, $\quad \boldsymbol{\eta} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d)$
    for $t = 0, \ldots, T-1$ **do**
        $\mathbf{a}_t \leftarrow \mathrm{softmax}(\beta_{\text{gen}}\,\mathbf{X}^\top\boldsymbol{\xi}_t)$
        $\boldsymbol{\epsilon}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d)$
        $\boldsymbol{\xi}_{t+1} \leftarrow (1-\alpha)\,\boldsymbol{\xi}_t + \alpha\,\mathbf{X}\,\mathbf{a}_t + \sqrt{2\alpha/\beta_{\text{gen}}}\,\boldsymbol{\epsilon}_t$
    Collect s evenly spaced samples from $\{\boldsymbol{\xi}_{T_b+1}, \boldsymbol{\xi}_{T_b+\Delta t+1}, \ldots, \boldsymbol{\xi}_T\}$ into $\mathcal{G}$
Decoding:
for each $\boldsymbol{\xi} \in \mathcal{G}$ **do**
    Inverse PCA: $\hat{\mathbf{x}} \leftarrow \bar{\mathbf{x}} + \mathbf{W}_d\,\boldsymbol{\xi} \in \mathbb{R}^{20L}$
    Argmax decode: $g_\ell \leftarrow \arg\max_{a\in\mathcal{A}}\ \hat{x}_{20(\ell-1)+a}$ for $\ell = 1, \ldots, L$
return $\{\mathbf{g}_1, \ldots, \mathbf{g}_S\}$

Temperature Selection and Sequence Decoding. The critical inverse temperature β^* is identified by the entropy inflection criterion used in prior stochastic-attention work. For each of $K_p = \min(K, 20)$ stored-pattern probes, $\boldsymbol{\xi} = \mathbf{m}_i$, we compute attention weights $\mathbf{w}_i(\beta) = \mathrm{softmax}(\beta\,\mathbf{X}^\top \mathbf{m}_i)$ and entropy $H_i(\beta) = -\sum_{k=1}^{K} w_{ik}(\beta) \log w_{ik}(\beta)$. We then average over probes, $\bar{H}(\beta) = K_p^{-1} \sum_i H_i(\beta)$, and locate the entropy inflection point on 50 logarithmically spaced values over $\beta \in [0.1, 500]$. Small β gives nearly uniform attention over memories, whereas increasing β produces pattern-selective attention basins. Generation uses $\beta_{\text{gen}} = \max(\lceil 2\beta^* \rceil, 5)$ to operate just above this transition while retaining stochastic exploration.

To decode a PCA-space sample $\boldsymbol{\xi}$ to an amino acid sequence, the PCA projection is inverted to recover a one-hot-like vector in the original $20L$-dimensional space, and the amino acid at each position is assigned by argmax over the 20 channels. The resulting length-L sequence is the decoded output.

Experimental Design. We selected eight Pfam families spanning a range of sizes, sequence lengths, and conservation levels (Table 1): RRM (PF00076, K=68, L=71), SH3 (PF00018, K=55, L=48), WW (PF00397, K=420, L=31), Kunitz (PF00014, K=99, L=53), zf-C2H2 (PF00096, K=151, L=23), PDZ (PF00595, K=44, L=83), Pkinase (PF00069, K=37, L=262), and Defensin_beta (PF00711, K=45, L=36). Seed alignments were downloaded from InterPro (30) and cleaned by removing columns with $> 50\%$ gaps and sequences with $> 30\%$ gaps. The principal component analysis projection retained 95% of variance, followed by unit-norm projection to form the memory matrix. For each family, we ran 30 independent ULA chains of T=5,000 iterations (α=0.01), discarding 2,000 burn-in iterations and thinning every 100 steps to retain 5 samples per chain (S=150 sequences total). The inverse temperature was set automatically via the entropy inflection criterion, with a generation regime at $\beta_{\text{gen}} = \max(\lceil 2\beta^* \rceil, 5)$ and a retrieval regime at $\beta_{\text{ret}} \approx 20\beta^*$. The thinning interval of 100 steps is well-matched to the integrated autocorrelation time of the Hopfield energy (τ_{int} = 76–122 steps across families), yielding approximately 35 effective independent samples per chain and ≈1,050 effective draws total; the burn-in of 2,000 steps provides a 3.2–6.6× margin over the empirical chain convergence point (SI Appendix, Table S11). Initialization sensitivity was verified by repeating the full pipeline with random sphere initialization ($\boldsymbol{\xi}_0 = \mathbf{z}/\|\mathbf{z}\|_2$, $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d)$) in place of stored-pattern initialization; output metrics were indistinguishable across all eight families (maximum absolute KL difference 0.0003), indicating that the generated ensemble is not an artifact of starting conditions (SI Appendix, Table S12).

To contextualize SA against methods that require pretraining on large external corpora, we compared four established baselines: (i) profile HMM emit via HMMER3 (5); (ii) EvoDiff MSA-conditioned diffusion (15), pretrained on UniRef50 MSAs; (iii) MSA Transformer (14) via iterative Gibbs-like masked language model sampling; and (iv) a Potts model fitted via pseudo-likelihood maximization (plmDCA) (8). For each baseline, the same 150-sequence budget and evaluation metrics (KL divergence, novelty, sequence identity) were applied. Because EvoDiff and the MSA Transformer emit variable-length, unaligned sequences, each baseline sequence was conformed to the alignment length L before metric computation by truncating longer sequences to L and padding shorter ones with alanine; this length normalization applies only to the variable-length baselines, not to SA, whose decoded sequences are already length L. Structural plausibility was assessed by predicting folded structures for 50 sequences per method per family using ESMFold (21) and AlphaFold2 (1) (via ColabFold (31)), comparing each predicted structure to an experimentally determined reference using TM-align (32) (reference structures listed in SI Appendix, Table S4). All statistical comparisons used Wilcoxon rank-sum tests with Benjamini-Hochberg false discovery rate (FDR) correction at q=0.05.

Acknowledgments. The idea for SA arose while developing a lecture on energy-based methods for eCornell, an opportunity without which this work would not have begun. I am deeply grateful to eCornell for making it possible. We thank the Pfam/InterPro consortium for maintaining publicly available seed alignments. Structure predictions were performed using ESMFold and AlphaFold2 (via ColabFold). Computations were performed on resources at Cornell University. This research received no external funding.

Data, Materials, and Software Availability

All code, data, seed alignments, generated sequences, baseline outputs, structure metrics and predictions, deep mutational scanning mappings, and analysis scripts needed to regenerate every table and figure are openly available. The exact version used for this study is archived at Zenodo (DOI: `[to be assigned on deposit]`), with ongoing development at https://github.com/varnerlab/SA-Protein-Modeling-Study. Pfam seed alignments were obtained from InterPro (30). No new experimental data were generated for this study; all validation used publicly available computational tools (ESMFold, AlphaFold2 via ColabFold, ESM2-650M, the MSA Transformer, EvoDiff, HMMER, and TM-align) and published deep mutational scanning datasets.

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Table 1. Properties of the eight Pfam families used in this study. K: number of seed sequences after cleaning. L: alignment length. d: PCA dimension (95% variance). $\bar{H}_{\mathrm{col}}$: mean per-column Shannon entropy (nats). K_{eff}: effective number of sequences at 80% identity. β^*: empirical entropy inflection point.

Family	Pfam ID	K	L	d	$\bar{H}_{\mathrm{col}}$	K_{eff}	β^*	$\beta^*/\sqrt{d}$
RRM	PF00076	68	71	59	1.92	68	3.85	0.50
SH3	PF00018	55	48	46	1.68	55	3.85	0.57
WW	PF00397	420	31	186	1.74	419	5.45	0.40
Kunitz	PF00014	99	53	80	1.61	99	3.85	0.43
zf-C2H2	PF00096	151	23	106	1.91	151	4.58	0.44
PDZ	PF00595	44	83	37	1.88	43	3.23	0.53
Pkinase	PF00069	37	262	34	1.72	37	3.23	0.55
Defensin_beta	PF00711	45	36	37	1.65	45	3.23	0.53

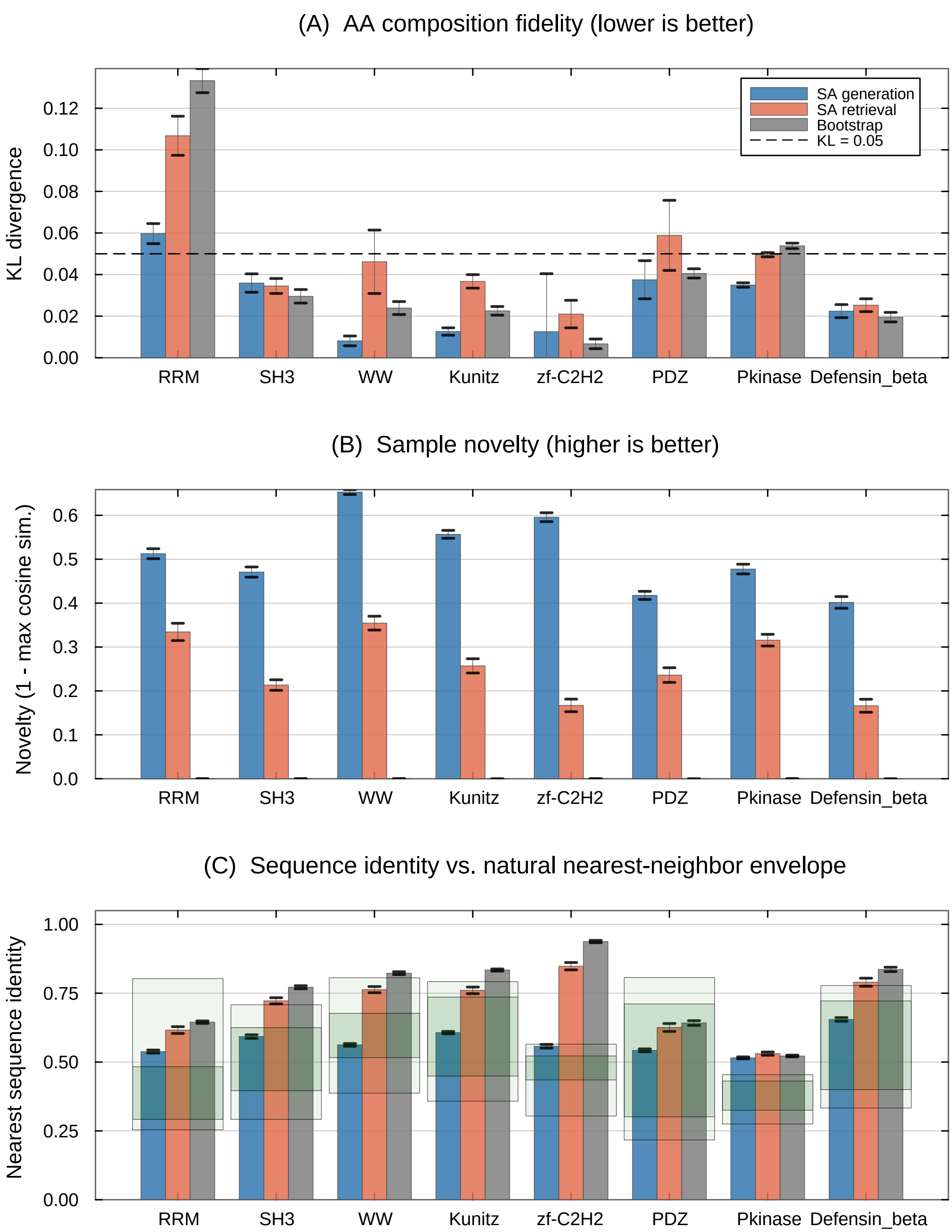

Fig. 1. Cross-family comparison of SA generation, SA retrieval, and bootstrap across eight Pfam families. **(A)** KL divergence of amino acid composition (lower is better); dashed line at KL$=0.05$. **(B)** Novelty in PCA space $(1 - \max_k \cos(\hat{\xi}, \mathbf{m}_k))$; higher is better). Bootstrap samples have zero novelty by construction. **(C)** Nearest sequence identity to a stored pattern; the green boxes show each family's natural within-family nearest-neighbor identity envelope (dark: 10th–90th percentile; light: full min–max range; SI Appendix, Table S14). The SA-generated identity lies within the envelope in seven of eight families; in Pkinase it rises above the natural maximum. Error bars: ± 1 SE (30 chains $\times$ 5 samples).

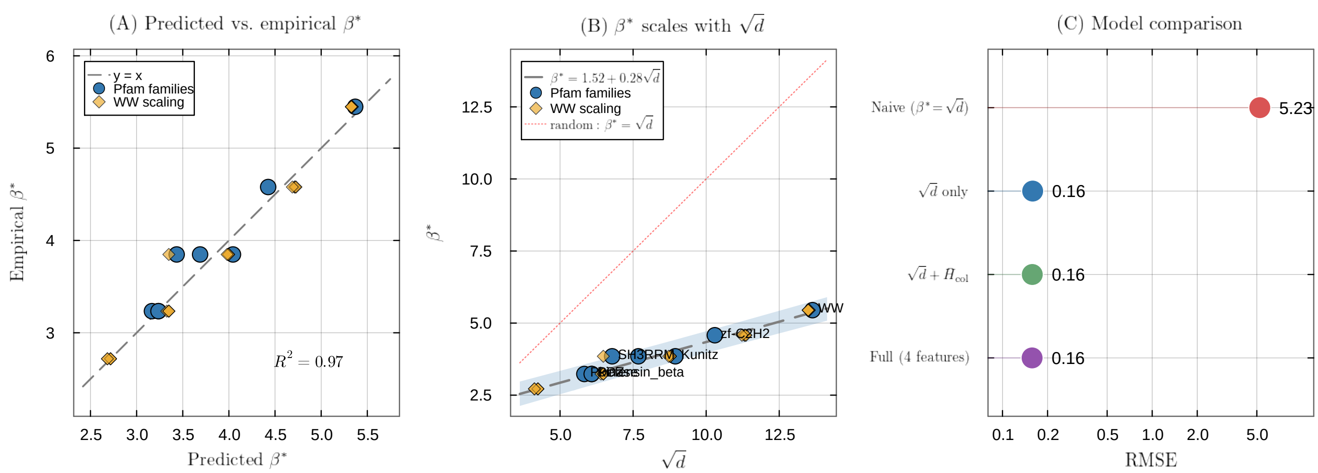

Fig. 2. Predicting the critical temperature from MSA statistics. **(A)** Predicted vs. empirical β^* for the simple linear model $\beta^* \approx 1.52 + 0.28\sqrt{d}$ (R^2=0.97, n=33). Blue circles: eight Pfam families; gold diamonds: WW domain scaling replicates ($K \in \{20, \dots, 400\}$, five repeats each). **(B)** β^* as a function of $\sqrt{d}$, with the fitted regression (dashed) and the unit-coefficient reference $\beta^* = \sqrt{d}$ (dotted red). The fitted intercept and reduced slope reflect structured correlations in real protein families. **(C)** Model comparison (log-scale RMSE). The fitted two-parameter model (RMSE=0.16) is $33\times$ more accurate than the unit-coefficient reference $\beta^* = \sqrt{d}$ (RMSE=5.23); adding MSA features provides negligible improvement.

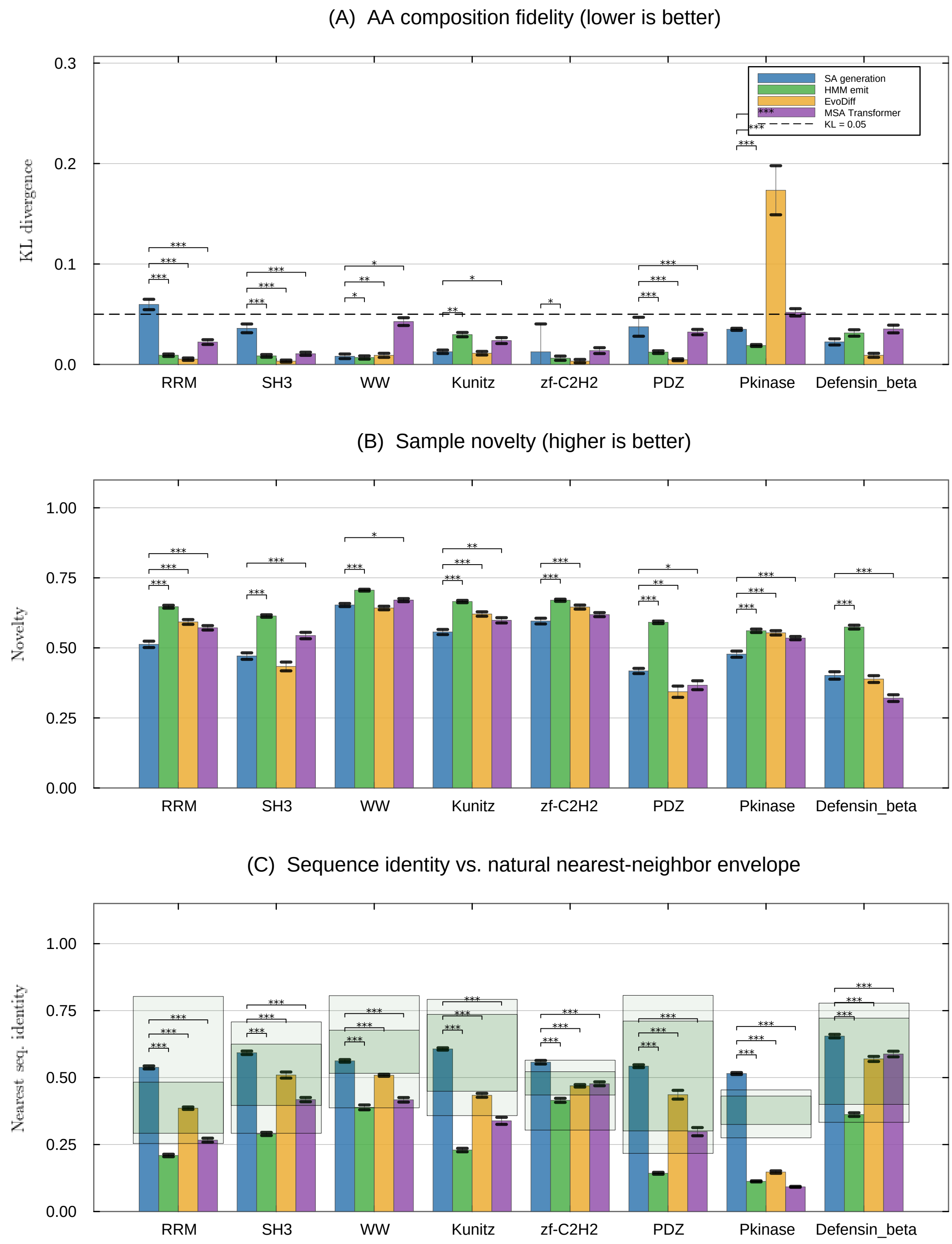

Fig. 3. Head-to-head comparison of SA generation against three learned baselines across all eight Pfam families. **(A)** KL divergence of amino acid composition (bootstrap SE; $n_{\text{boot}}=1000$); dashed line at KL$=0.05$. **(B)** Novelty in PCA space. **(C)** Nearest sequence identity to a stored pattern; the green boxes show each family's natural within-family nearest-neighbor identity envelope (dark: 10th–90th percentile; light: full min–max range; SI Appendix, Table S14). Panels (B) and (C): per-chain means ± 1 SE. Significance brackets: Wilcoxon rank-sum tests ($^{***}p < 0.001$, $^{**}p < 0.01$, $^{*}p < 0.05$). Per-family numerical values are reported in SI Appendix for the learned baselines (Table S2) and for SA generation and the reference conditions (Table S1).

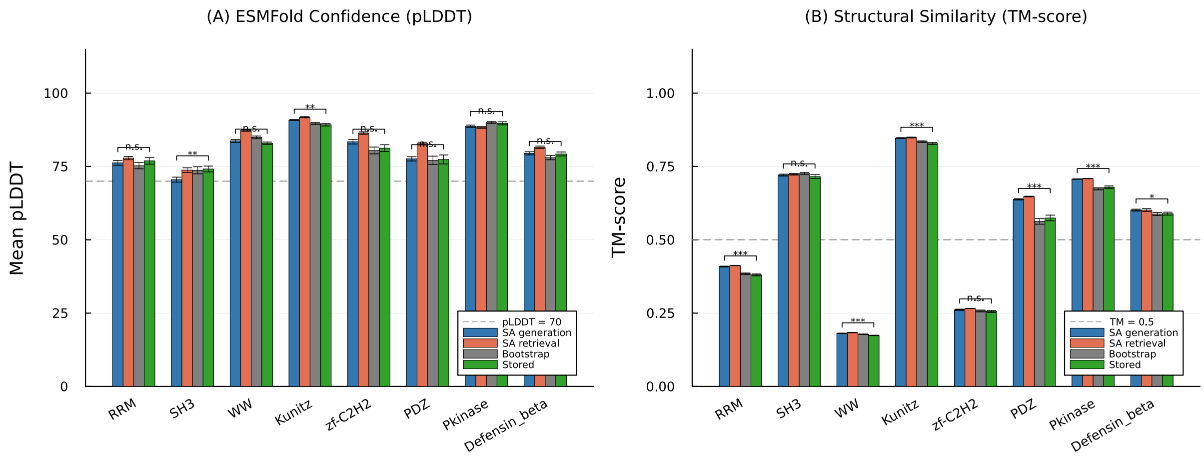

Fig. 4. Structure validation using ESMFold. **(A)** Mean pLDDT across all eight families for SA generation, SA retrieval, bootstrap, and stored sequences; dashed line at pLDDT$=70$. **(B)** TM-score to the experimentally determined reference structure; dashed line at TM$=0.5$. Significance: Wilcoxon rank-sum tests ($^{***}p < 0.001$, $^{**}p < 0.01$, $^{*}p < 0.05$, n.s. = not significant). Error bars: ± 1 SE (n=50).

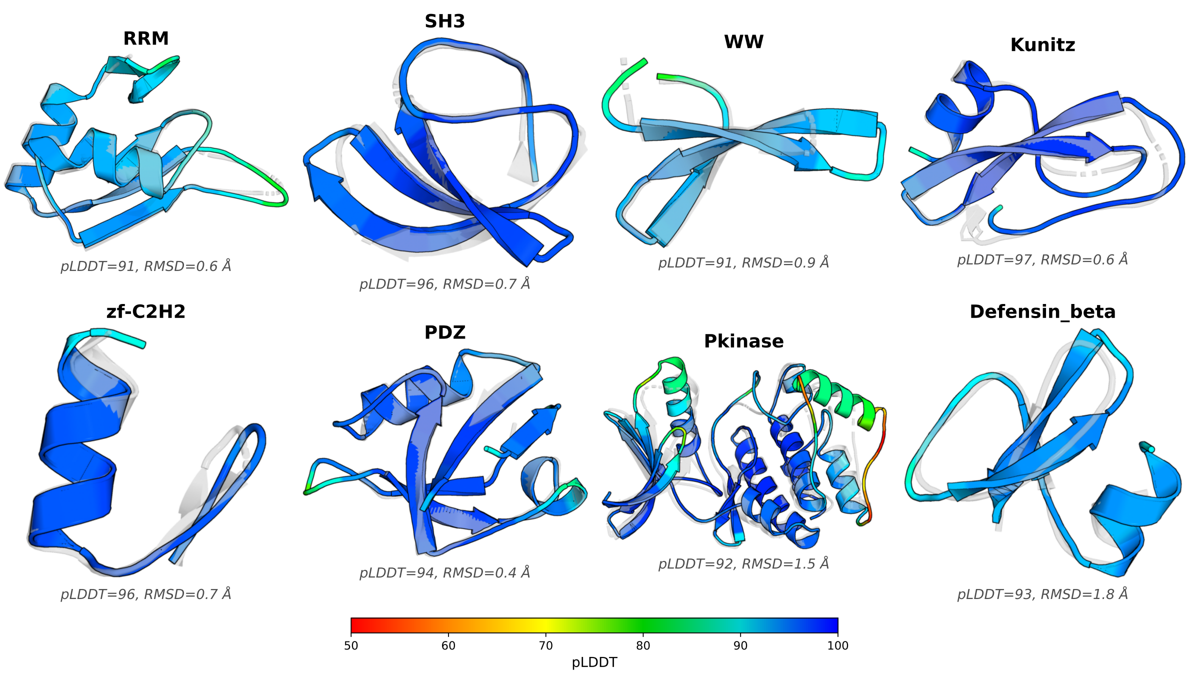

Fig. 5. AlphaFold2 predicted structures for representative SA-generated sequences (colored by per-residue pLDDT confidence) superimposed on the experimentally determined reference structure for each family (gray, semi-transparent). For visualization, the SA-generated sequence with the highest mean pLDDT in each family was selected and aligned to the reference using PyMOL; aggregate structural statistics for the generated ensembles are reported in Fig. 4 and SI Appendix, Figs. S3–S4. The close predicted structural agreement (sub-angstrom root-mean-square deviation (RMSD)) and uniformly high pLDDT scores are consistent with SA-generated sequences encoding three-dimensional folds that recapitulate the canonical architecture of their target family, despite 40–65% sequence novelty.